

Minimising total chemo treatment time

The problem can be formulated as a linear program: minimize the total time $T = x_1 + x_2$ subject to the dosage constraint $x_1 + a_2 x_2 = 1$ with $x_1, x_2 \geq 0$. By general linear-programming theory, the feasible set is a line segment and the optimum occurs at one of its vertices ¹ ². These vertices correspond to using only the first therapy or only the second. Since the first-rate $a_1=1$ exceeds a_2 (the code's sum of powers of $1/2$ makes a_2 just under 1), using only the first therapy yields a smaller total time.

Concretely, the two corner solutions are $(x_1, x_2) = (1, 0)$ (giving dosage $1 \cdot 1 + 0 \cdot a_2 = 1$) and $(x_1, x_2) = (0, 1/a_2)$ (giving $0 + (1/a_2)a_2 = 1$). Writing the objective as $1 + (1-a_2)x_2$ shows it increases with x_2 when $a_2 < 1$. Thus the minimum is attained at $x_2=0, x_1=1$. In two-decimal accuracy (ℓ^∞ error < 0.01) this gives $x_1=1.00, x_2=0.00$, which indeed delivers total dosage 1 in the minimal time (1 second) ².

Answer: $x_1 \approx 1.00, x_2 \approx 0.00$ (to two decimal places).

¹ Linear programming - Wikipedia

https://en.wikipedia.org/wiki/Linear_programming

² sites.math.washington.edu

<https://sites.math.washington.edu/~perkins/381Awin14/handouts/chapter3.pdf>